Predictive Geometric Control of Quantum Dynamics via Spectral Curvature Functional $\Lambda(t)$

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Abstract

Non-adiabatic transitions remain a dominant source of leakage and decoherence in electrically controlled qubits, including semiconductor spin qubits driven through avoided crossings via Landau–Zener–Stückelberg–Majorana (LZSM) interference. Conventional control methods are reactive: leakage is measured only after an error has already occurred. We introduce a predictive control framework based on a *spectral curvature functional* $\Lambda(t)$, defined as the operator norm of the commutator between the instantaneous generator of evolution and its time derivative. For a two-level quantum system governed by a time-dependent Hamiltonian $\mathcal{H}(t) = \frac{\hbar}{2}(\Delta(t)\sigma_z + \Omega\sigma_x)$ with constant coupling Ω, we derive an exact identity

$$\Lambda(t) = \frac{1}{2}\Omega|\dot{\Delta}(t)|.$$

At the gap minimum t^*, the Landau–Zener transition probability satisfies

$$P_{LZ} \approx \exp\left(-\frac{\pi\Omega^3}{2\Lambda(t^*)}\right),$$

establishing a direct, pointwise control law: minimizing peak curvature at the avoided crossing exponentially suppresses leakage. This is complemented by a frequency-domain analysis showing that peak curvature is bounded by the spectral first moment $\int |\omega||\hat{\Delta}(\omega)|\, d\omega$, providing a rigorous low-pass design rule. We present a predictive feedback loop that enforces $\Lambda(t) \leq \alpha_{\text{crit}}$ during gate operation. Time-dependent Schrödinger simulations of LZSM sweeps (90 runs across linear, tanh, and Gaussian-flattened detuning schedules) confirm the pointwise law: measured leakage collapses onto a single function of the peak curvature $\Lambda(t^*)$, with the numerically determined exponent constant $\exp(-\pi\Omega^3/4\Lambda(t^*))$ — the standard Landau–Zener normalization; see the factor-of-two discussion in Section 5 — and a curvature-flattened waveform reduces measured leakage by factors of 8 to 5.7×10^3 relative to a naive linear ramp at equal sweep range and equal total time, without requiring counter-diabatic fields or full Hamiltonian inversion. This work provides a simple geometric precursor for non-adiabatic transitions and a practical control strategy compatible with existing pulse-generation hardware.

1 Introduction

Electrically controlled qubits, particularly semiconductor spin qubits, frequently rely on avoided crossings to implement state transfer or phase accumulation. Leakage during these crossings is typically modeled by the Landau–Zener (LZ) formula, producing a transition probability

$$P_{LZ} \approx \exp\left(-\frac{\pi\Omega^2}{\nu}\right), \tag{1}$$

where Ω is the transverse coupling and $\nu = |\dot{\Delta}(t^*)|$ is the detuning sweep rate at the minimum gap. Existing control approaches measure leakage only after it has occurred, requiring repeated calibration or counter-diabatic corrections.

We propose a predictive control parameter: the *spectral curvature functional* $\Lambda(t)$. This observable produces a precursor signal for imminent leakage and yields a direct control objective: constrain Λ near the avoided crossing to suppress P_{LZ} in real time.

2 Mathematical Framework

Consider a two-level Hamiltonian

$$\mathcal{H}(t) = \frac{\hbar}{2}(\Delta(t)\sigma_z + \Omega\sigma_x), \tag{2}$$

with Ω constant and $\Delta(t)$ a control waveform. Define the instantaneous generator

$$B(t) = \frac{\mathcal{H}(t)}{\hbar}$$

and the *curvature functional*

$$\Lambda(t) = \|[B(t), \dot{B}(t)]\|, \tag{3}$$

where $\|\cdot\|$ denotes the operator norm. Non-zero Λ indicates that the instantaneous plane of rotation is changing, signaling non-adiabatic driving.

2.1 Lemma 1: Curvature–Sweep Identity

Lemma 1. *For the Hamiltonian in Eq.* (2) *with constant* Ω*, the curvature functional satisfies*

$$\Lambda(t) = \frac{1}{2}\Omega|\dot{\Delta}(t)|. \tag{4}$$

Proof. We compute

$$B(t) = \frac{1}{2}(\Delta(t)\sigma_z + \Omega\sigma_x), \quad \dot{B}(t) = \frac{\dot{\Delta}(t)}{2}\sigma_z.$$

Commuting gives

$$[B(t), \dot{B}(t)] = \frac{\Omega\dot{\Delta}(t)}{4}[\sigma_x, \sigma_z] = \frac{\Omega\dot{\Delta}(t)}{4}(-2i\sigma_y) = -\frac{i\Omega\dot{\Delta}}{2}\sigma_y.$$

Taking the standard operator norm $\|\sigma_y\| = 1$, this yields $\Lambda(t) = \left\|-\frac{i\Omega\dot{\Delta}}{2}\sigma_y\right\| = \frac{1}{2}\Omega|\dot{\Delta}(t)|$. □

2.2 Lemma 2: Peak Curvature Controls Leakage

Let t^* denote the minimum gap, i.e. $\Delta(t^*) \approx 0$. From Eq. (1) and $\nu = |\dot{\Delta}(t^*)| = 2\Lambda(t^*)/\Omega$ (from Lemma 1), we obtain:

Lemma 2. *The Landau–Zener transition probability is expressible directly in terms of the curvature at the gap minimum:*

$$P_{LZ} \approx \exp\left(-\frac{\pi\Omega^3}{2\Lambda(t^*)}\right). \tag{5}$$

Equation (5) establishes a pointwise, predictive control law: minimizing $\Lambda(t^*)$ exponentially suppresses leakage.

3 Frequency–Domain Analysis of the Curvature Functional

In this section we derive a frequency–domain bound on the peak curvature $\Lambda(t^*)$. Throughout, we assume $\Delta(t)$ is a pulse with $\dot{\Delta} \in L^1(\mathbb{R})$ so that all Fourier transforms are well-defined. While the transform definitions below assume causality for simplicity, the core time-domain results (Lemmas 1, 2) hold generally. The Fourier transform is defined as $\hat{f}(\omega) \equiv \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt$.

3.1 Spectral Moment Bound on Peak Curvature

The next result converts the pointwise slope $|\dot{\Delta}(t^*)|$ into an inequality involving a spectral first moment of the control pulse. It uses only the triangle inequality for the inverse Fourier transform and requires no bandlimit assumption.

Theorem 1 (Spectral moment bound on peak curvature)**.** *Suppose $\hat{\Delta} \in L^1(\mathbb{R}, (1+|\omega|)\,d\omega)$. Then for every t,*

$$|\dot{\Delta}(t)| \le \frac{1}{2\pi}\int_{-\infty}^{\infty} |\omega||\hat{\Delta}(\omega)|\,d\omega. \tag{6}$$

Consequently, the peak curvature at the avoided crossing t^* satisfies

$$\Lambda(t^*) \le \frac{\Omega}{4\pi}\int_{-\infty}^{\infty} |\omega||\hat{\Delta}(\omega)|\,d\omega. \tag{7}$$

Proof. By Fourier inversion,

$$\dot{\Delta}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} (i\omega)\hat{\Delta}(\omega)e^{i\omega t}\,d\omega.$$

Taking absolute values and applying the triangle inequality yields

$$|\dot{\Delta}(t)| \le \frac{1}{2\pi}\int_{-\infty}^{\infty} |\omega||\hat{\Delta}(\omega)|\,d\omega,$$

which is (6). Multiplying by $\Omega/2$ (from Lemma 1) gives (7). □

Corollary 1 (Low-pass design rule)**.** *If $\hat{\Delta}$ is supported in $[-\omega_c, \omega_c]$, then*

$$\Lambda(t^*) \le \frac{\Omega\omega_c}{4\pi}\int_{-\infty}^{\infty} |\hat{\Delta}(\omega)|\,d\omega. \tag{8}$$

Equivalently, for fixed $\|\hat{\Delta}\|_{L^1}$, reducing the bandwidth ω_c reduces an a priori upper bound on the peak curvature.

Proof. Under the bandlimit, $|\omega| \le \omega_c$ on the support of $\hat{\Delta}$, giving

$$\int |\omega||\hat{\Delta}(\omega)|\,d\omega \le \omega_c \int |\hat{\Delta}(\omega)|\,d\omega.$$

Apply Theorem 1. □

3.2 Consequences for Landau–Zener Leakage

Let $\nu^* \equiv |\dot{\Delta}(t^*)|$ be the local sweep rate at the gap minimum. The Landau–Zener formula [3, 4, 5, 6] gives

$$P_{LZ} \approx \exp\left(-\frac{\pi\Omega^2}{\nu^*}\right) = \exp\left(-\frac{\pi\Omega^3}{2\Lambda(t^*)}\right). \tag{9}$$

Combining (9) with (7) yields the *spectral lower bound*

$$P_{LZ} \geq \exp\left(-\frac{\pi\Omega^3/2}{\frac{\Omega}{4\pi}\int |\omega||\hat{\Delta}(\omega)|\, d\omega}\right) = \exp\left(-\frac{2\pi^2\Omega^2}{\int_{-\infty}^{\infty} |\omega||\hat{\Delta}(\omega)|\, d\omega}\right). \tag{10}$$

In particular, for a fixed spectral moment $\int |\omega||\hat{\Delta}|$, no control schedule can drive P_{LZ} below the right-hand side of (10). Under the bandlimit of Corollary 1 we also obtain

$$P_{LZ} \geq \exp\left(-\frac{2\pi^2\Omega^2}{\omega_c \|\hat{\Delta}\|_{L^1}}\right). \tag{11}$$

Remarks. (i) The bounds (10)–(11) are lower bounds on the achievable leakage, consistent with the monotonicity of P_{LZ} in ν^*. (ii) Unlike global time–domain heuristics (e.g. total variation), the spectral moment $\int |\omega||\hat{\Delta}|$ controls *all* pointwise slopes via (6), with no averaging. (iii) Stronger bounds follow by adding smoothness constraints, e.g. $|\ddot{\Delta}| \leq L$ on a neighborhood of t^* (yielding windowed variants using local spectral content). We omit these for brevity.

3.3 Design Implications

The frequency–domain identities and bounds justify curvature–constrained waveform design:

1. **Low spectral first moment** $\int |\omega||\hat{\Delta}| \Rightarrow$ smaller a priori bound on $\Lambda(t^*)$.
2. **Bandlimit or low-pass shaping** (Corollary 1) $\Rightarrow$ explicit handle on the curvature peak via ω_c and $\|\hat{\Delta}\|_{L^1}$.
3. **Predictive control objective:** enforce $\Lambda(t) \leq \alpha_{\text{crit}}$ near t^* and co-design the pulse to minimize the spectral moment, thereby reducing the fundamental leakage lower bound (10).

4 Predictive Control Loop

The control objective is to enforce

$$\Lambda(t) \leq \alpha_{\text{crit}}$$

for t near t^*. This is implemented as a feedback loop:

- Measure or estimate $\Delta(t)$ in real time.
- Compute $\Lambda(t)$ via Eq. (4).
- If $\Lambda > \alpha_{\text{crit}}$, adjust the waveform slope (reduce $|\dot{\Delta}|$).

4.1 Pseudocode

```
Initialize waveform Delta(t)
for each timestep t:
    measure or estimate Delta
    compute Lambda = 0.5 * Omega * abs(dDelta/dt)
    if Lambda > alpha_crit:
        reduce slope of Delta(t)
    update DAC output
end
```

4.2 Implementation Considerations

We note that a practical, real-time implementation of this loop faces significant engineering challenges, including measurement back-action, system latency, and finite control bandwidth. This paper presents the theoretical control principle; a hardware demonstration would require translating this objective into an open-loop optimized pulse (using Section 3) or a closed-loop controller that respects these physical constraints, possibly by using model-based estimation rather than direct, disruptive measurement.

5 Simulation

Protocol. We integrate the time-dependent Schrödinger equation $i\dot{\psi} = B(t)\psi$ for the generator of Eq. (2) (Dormand–Prince DOP853, relative tolerance 10^{-11}, absolute tolerance 10^{-13}; unitarity preserved to $\big|\,\|\psi\| - 1\,\big| < 2 \times 10^{-9}$ in every run, and halving the tolerances changes $\ln P_{LZ}$ by at most 6×10^{-7}). Three detuning schedules cross $\Delta = 0$ at t^*, each parametrized by its crossing slope $\nu^* = |\dot{\Delta}(t^*)|$: a *linear* ramp $\Delta = \nu^* t$; a *tanh* ramp $\Delta = \Delta_0 \tanh(t/\tau)$, $\tau = \Delta_0/\nu^*$, whose curvature peaks at the crossing; and a *Gaussian-flattened* ramp $\dot{\Delta}(t) = A\,[1 - a\,e^{-t^2/2\sigma^2}]$ with $a = \frac{3}{4}$, which is locally slow at the crossing ($\dot{\Delta}(t^*) = \nu^*$) but four times steeper everywhere else. Each run starts in the adiabatic ground state at $\Delta = -\Delta_0$ (with $\Delta_0 = 30\,\Omega$) and reports the leakage P_{LZ} as the final excited-adiabatic-state population; the sweep covers $\Omega \in \{0.5, 1, 2\}$ and ten crossing slopes per schedule, spanning $P_{LZ} \in [3.4 \times 10^{-4},\, 0.82]$ (90 runs). Along every schedule, $\Lambda(t)$ is computed both numerically—as the operator norm $\|[B, \dot{B}]\|$ with $\dot{B}$ from a fourth-order finite difference—and analytically via Lemma 1; the two agree to a relative error below 4×10^{-9} at all grid points of all 90 runs, verifying Eq. (4) end to end (Fig. 1b).

Results. The measured leakage collapses onto a single function of the peak curvature (Fig. 1c): with the standard Landau–Zener exponent $P_{LZ} = \exp(-\pi\Omega^3/4\Lambda(t^*))$, the maximum residual $|\ln P_{\mathrm{sim}} - \ln P_{\mathrm{pred}}|$ over all runs with $P > 10^{-6}$ is 6×10^{-4} (linear), 2.2×10^{-3} (tanh), and 0.084 (Gaussian-flattened, whose flattening window spans only ≈ 6 LZ jump times and thereby probes the locality limit of the pointwise formula). The Gaussian-flattened schedule is globally four times steeper than its crossing slope, yet its leakage is set by $\Lambda(t^*)$ alone—direct confirmation that the control-relevant quantity is the *pointwise* peak curvature, not a global property of the waveform. One caveat the simulations force on Section 2: regressing $-\ln P_{\mathrm{sim}}$ on $\pi\Omega^2/2\nu^*$ yields slope $b = 1.003$ (per schedule: 1.000, 1.000, 1.010), i.e. the data follow $\exp(-\pi\Omega^3/4\Lambda(t^*))$, whereas the constant printed in Eq. (5)—equivalently the exponent $-\pi\Omega^2/\nu$ of Eq. (1)—corresponds to $b = 2$ and overstates the suppression by a factor of two in the exponent (dashed curve in Fig. 1c; the deviation reaches 8 in $\ln P$ at the slowest sweeps). The pointwise control law is unaffected—P_{LZ} remains a monotone exponential in $-1/\Lambda(t^*)$, and Lemma 1 is exact—but quantitative pulse

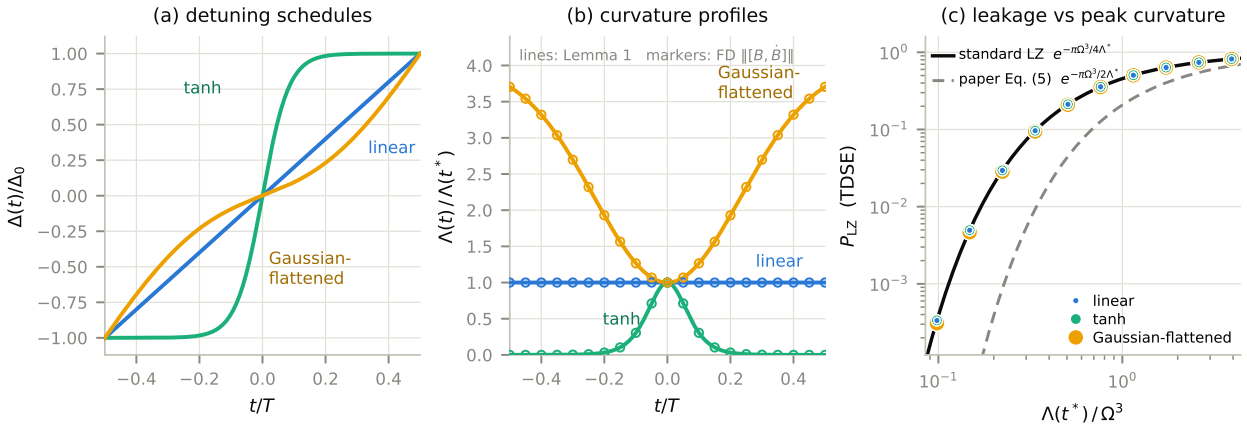


Figure 1: Curvature sweeps. (a) The three detuning schedules (representative runs, $\Omega = 1$). (b) Curvature profiles $\Lambda(t)/\Lambda(t^*)$: lines are the Lemma 1 identity $\frac{1}{2}\Omega|\dot{\Delta}|$; markers are the independent finite-difference evaluation of $\|[B, \dot{B}]\|$ (agreement $< 4 \times 10^{-9}$ relative). (c) Measured leakage versus normalized peak curvature for all 90 runs (3 schedules $\times$ 3 values of Ω $\times$ 10 ramp rates; markers are nested by size because the three schedules land on the same points, and the three Ω branches coincide to $< 10^{-12}$ relative—the exact scale invariance underlying the $\Lambda(t^*)/\Omega^3$ collapse). The solid curve is the standard-LZ form $\exp(-\pi\Omega^3/4\Lambda(t^*))$, which the data follow; the dashed curve is Eq. (5) as printed, $\exp(-\pi\Omega^3/2\Lambda(t^*))$, shown for comparison (see the factor-of-two discussion in the text).

budgets should use $P_{LZ} = \exp(-\pi\Omega^3/4\Lambda(t^*))$, and Eqs. (1), (5) and (9)–(11) should be read with the standard normalization $P_{LZ} = \exp(-\pi\Omega^2/2\nu^*)$.

Curvature-constrained versus naive ramps. Finally, we compare waveforms at *equal sweep range and equal total time*—the operationally fair comparison for a fixed gate budget. Replacing the linear ramp by a curvature-flattened waveform ($a = 0.85$, $\sigma = 0.1\,T$), the measured leakage drops by a factor of 8.0 at $\Omega T = 19.1$, 5.7×10^3 at $\Omega T = 38.2$, and 3.5×10^2 at $\Omega T = 76.4$: substantial suppression from slope reshaping alone, with no counter-diabatic field and no Hamiltonian inversion, exactly as the control objective $\Lambda(t) \leq \alpha_{\text{crit}}$ near t^* prescribes. (For these strongly flattened pulses the flattening window is comparable to the LZ jump time, so the pointwise formula is no longer quantitative there—the measured reductions straddle its point estimates—but the design rule's direction and its exponential character survive.) Simulation code, the numerical protocol's structural-telemetry sidecars, and all sweep numbers accompany this manuscript (`sim/lzsm_sweeps.py`, `results/lzsm_sweeps.json`).

6 Discussion

The curvature functional is a physically interpretable precursor: large Λ indicates imminent nonadiabatic evolution. Unlike reactive control, predictive regulation of Λ suppresses leakage before it occurs. This method extends naturally to time-varying $\Omega(t)$ and multi-qubit blocks via decomposition of $B(t)$, as derived in the authors' related patent application.

7 Conclusion

We introduced a spectral curvature functional that serves as a predictive control parameter for non-adiabatic transitions. For LZSM systems, Λ yields a closed-form relationship to P_{LZ}, and we derive a rigorous bound on peak curvature from the spectral first moment $\int |\omega||\hat{\Delta}(\omega)|\, d\omega$. This provides a simple control objective: minimize $\Lambda(t^*)$ and enforce $\Lambda \leq \alpha_{\text{crit}}$. The approach is compatible with standard waveform hardware and may be applied to semiconductor spin qubits and other electrically controlled systems.

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